

## Assignment

### module 2 : Installation and Maintenance of Hardware and Its

#### *Section 1: Multiple Choice*

1. Which of the following precautions should be taken before working on computer hardware?

Ans: (b) Wear an anti-static wrist strap to prevent damage from electrostatic discharge.

2. What is the purpose of thermal paste during CPU installation?

Ans: (c) To improve thermal conductivity between the CPU and the heat sink.

3. Which tool is used to measure the output voltage of a power supply unit (PSU)?

Ans: (a) Multimeter

4. Which component is responsible for storing BIOS settings, such as date and time, even when the computer is powered off?

Ans: (a) CMOS battery

#### *Section 2: True or False*

5. When installing a new hard drive, it is essential to format it before use.

Ans: True

6. A POST (Power-On Self-Test) error indicates a problem with the CPU.

Ans: False

7. True or False: It is safe to remove a USB flash drive from a computer without ejecting it first.

Ans: False

### *Section 3: Short Answer*

8. Describe the steps involved in installing a new graphics card in a desktop computer.

Ans: ☐ Steps to Install GPU

1. Shut down PC and unplug power.
2. Open case and wear anti-static strap.
3. Locate PCIe x16 slot and remove slot cover.
4. Insert graphics card firmly and screw it in place.
5. Connect PSU power cables (if needed).
6. Close case and reconnect monitor to GPU.
7. Power on PC and install latest drivers.

9. What is RAID, and what are some common RAID configurations?

Ans; RAID (Redundant Array of Independent/Inexpensive Disks) is a technology that combines multiple hard drives or SSDs into a single logical unit to improve performance, reliability, or both.

#### □ **Common RAID Configurations**

##### **1. RAID 0 (Striping)**

- Data split across 2 or more drives.
- ✓ Fast performance.
- ✗ No redundancy (if one drive fails, all data is lost).

##### **2. RAID 1 (Mirroring)**

- Data is duplicated on 2 drives.
- ✓ High reliability, data safe if one drive fails.
- ✗ Storage capacity is cut in half.

##### **3. RAID 5 (Striping with Parity)**

- Needs at least 3 drives.
- Data + parity (error checking) spread across all drives.
- ✓ Good balance of performance, storage, and redundancy.
- ✗ Slower write speeds compared to RAID 0.

##### **4. RAID 10 (1+0, Mirroring + Striping)**

- Needs at least 4 drives.
- Combines RAID 1 and RAID 0 (speed + redundancy).
- ✓ Very fast and reliable.
- ✗ Uses more drives, higher cost.

#### *Section 4: Practical Application*

10. Demonstrate how to replace a CPU fan in a desktop computer.

Ans: □ Steps to Replace a CPU Fan

Power off PC and unplug.

- Open case, wear anti-static strap.
- Locate CPU fan on motherboard.
- Unplug old fan connector.
- Unscrew/unclip and remove old fan.
- Clean dust / old thermal paste.
- Install new fan and apply paste if needed.
- Plug fan into CPU\_FAN header.
- Close case, power on, test fan.

### *Section 5: Essay*

11. Discuss the importance of regular maintenance for computer hardware and provide examples of maintenance tasks.

Ans: □ Importance of Regular Maintenance for Computer Hardware

Regular maintenance keeps a computer **fast, reliable, and long-lasting**. It helps prevent failures, improves performance, and reduces costly repairs. Without it, dust, heat, and wear can damage components.

## □ **Examples of Maintenance Tasks**

1. **Cleaning Dust** – Remove dust from fans, vents, and components to prevent overheating.
2. **Checking Cables & Connections** – Ensure power and data cables are properly seated.
3. **Updating Software & Drivers** – Keeps hardware running smoothly and securely.
4. **Running Disk Cleanup/Defrag (HDDs)** – Improves storage performance.
5. **Replacing Thermal Paste** – Maintains good heat transfer between CPU and cooler.
6. **Monitoring Hardware Health** – Check temperatures, fan speeds, and storage health (SMART tools).
7. **Backup Data** – Protects against hardware failure or crashes.